Static and kinetic friction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish adjustable static friction from sliding friction.

SUCCESS CRITERIA - I can...

1. Compare a push with the static-friction limit.
2. Find actual friction for rest or sliding.
3. Explain friction direction using relative slipping.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A spacecraft of mass 3 kg pushes an asteroid of mass 300 kg with 15 N. Compare interaction forces and the accelerations due to that interaction.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $0 \le f_s \le \mu_s N$; $f_k = \mu_k N$.
s = static; k = kinetic; N = normal force. Static friction adjusts up to its maximum. Kinetic friction applies during sliding.

Worked reasoning: A box remains still while pushed with 3 N. Static friction is 3 N opposite the push, provided its maximum is at least 3 N. Static friction need not equal its maximum.

Your turn: Draw the horizontal forces. Then redraw for a 5 N push that is still below the maximum.

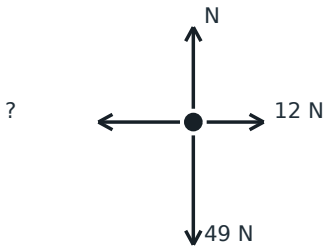


Box remains at rest. Redraw for a 5 N push.

Marzano check - circle: ① ② ③

03 Actual versus maximum

A 5 kg box is initially at rest on a level floor. Its friction coefficients are $\mu_s = 0.40$ and $\mu_k = 0.30$. A horizontal push is 12 N. Use g = 9.8 N/kg. Find the normal force, maximum static friction, actual friction and acceleration. Label the diagram.



The dot represents the object. Initially at rest

Marzano check - circle: ① ② ③

04 Sliding case

Increase the push on that box to 25 N. Find friction and acceleration after it slides.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Estimate from data

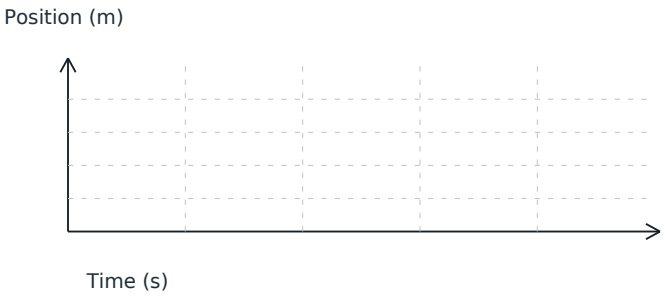
A class tests a block with a 20 N normal force. Use the repeated measurements to find mean peak force, mean sliding force, and estimates of μ_s and μ_k. Which estimate describes the threshold for slipping?

Trial	Peak force (N)	Sliding force (N)
1	8	6
2	9	5
3	7	7

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

x_i=-4±0.5 m, v=6±0.1 m/s and exact t=8 s. Give central final position, its conservative interval and full width. Name an assumption.
Label the model with the given positions and any prediction.



Marzano check - circle: ① ② ③

07 Exit check

A 10 kg box has μ_s = 0.50 and μ_k = 0.40. Use g = 9.8 N/kg. Compare a 30 N push with a 60 N push, starting from rest each time. Find actual friction and acceleration in both cases.

Push (N)	Still or sliding?	Friction (N)	a (m/s²)
30	___	___	___
60	___	___	___

Marzano check - circle: ① ② ③

08 STRETCH: test the model

A 5 kg box has μ_s=0.4 and μ_k=0.3 on a level floor; use g=10 N/kg. A horizontal 22 N pull begins from rest. Find maximum static friction, acceleration after motion starts, and the pull needed for steady sliding.



Add and label force arrows.

Draw the sliding case, with actual friction.

Marzano check - circle: ① ② ③